

Spark Exercises & Analysis

1. CUSTOMER DATA

Doc Analysis

Format	Time (sec)	Jobs	Stages	Tasks	Shuffle Size	Notes
CSV	16.6	1	1	50	~148 KB	Fast, simple text output
JSON	15.4	1	1	50	~148 KB	Fastest in this run
Parquet	17.0	1	2	50	Small	Columnar format, extra metadata

Console

```
Terminal Local x +
25/12/02 16:55:31 WARN WadlFeature: JAXSContext implementation could not be found. WADL feature is disabled.
25/12/02 16:55:38 INFO CustomersCityCount$: CSV WRITE DONE.
Execution Time: 70605 ms
25/12/02 16:56:52 INFO CustomersCityCount$: JSON WRITE DONE.
Execution Time: 73049 ms
25/12/02 16:58:08 INFO CustomersCityCount$: PARQUET WRITE DONE.
Execution Time: 75789 ms
Execution Time: 223218 ms
25/12/02 16:58:08 INFO CustomersCityCount$: COMPLETED.
[success] Total time: 246 s (0:04:06.0), completed 02-Dec-2025, 4:58:08 PM
```

Executors

spark 3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

CustomersCityCount application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	99.8 KiB / 434.4 MiB	0.0 B	11	0	0	208	208	8.3 min (3 s)	0.0 B	461 KiB	553.8 KiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	99.8 KiB / 434.4 MiB	0.0 B	11	0	0	208	208	8.3 min (3 s)	0.0 B	461 KiB	553.8 KiB	0

Executors

Show 20 entries

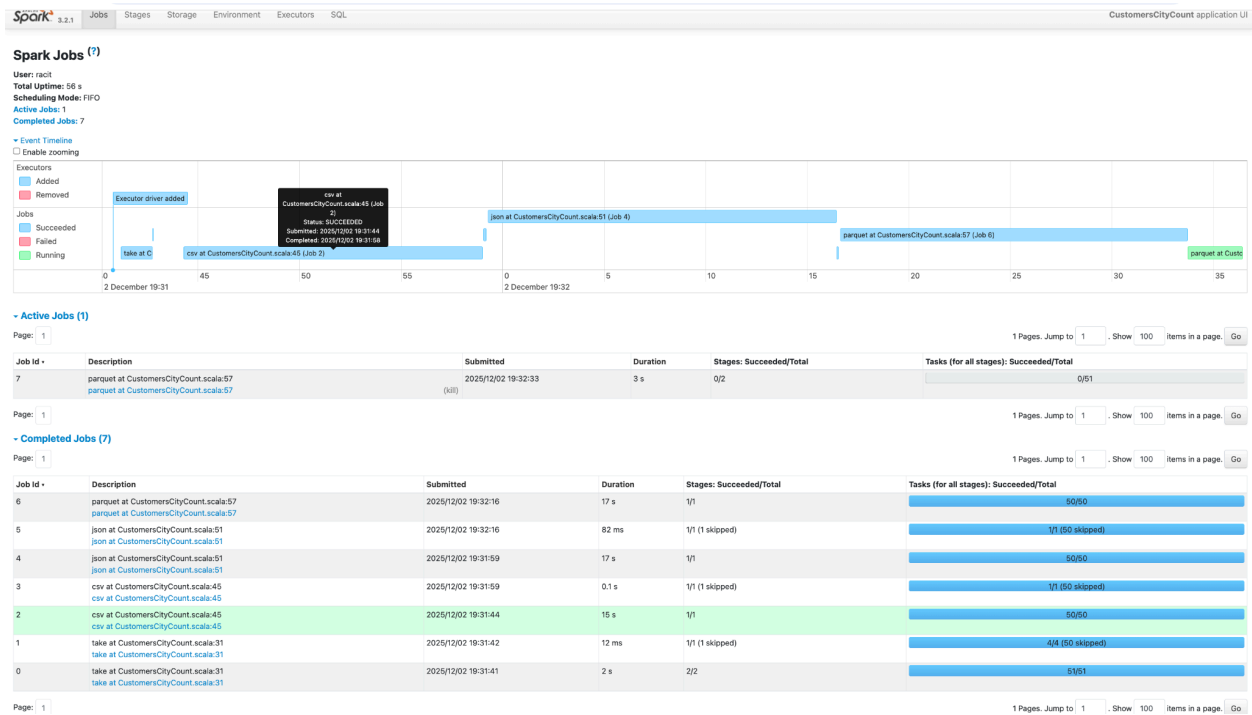
Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.23-58573	Active	0	99.8 KiB / 434.4 MiB	0.0 B	11	0	0	208	208	8.3 min (3 s)	0.0 B	461 KiB	553.8 KiB	Thread Dump

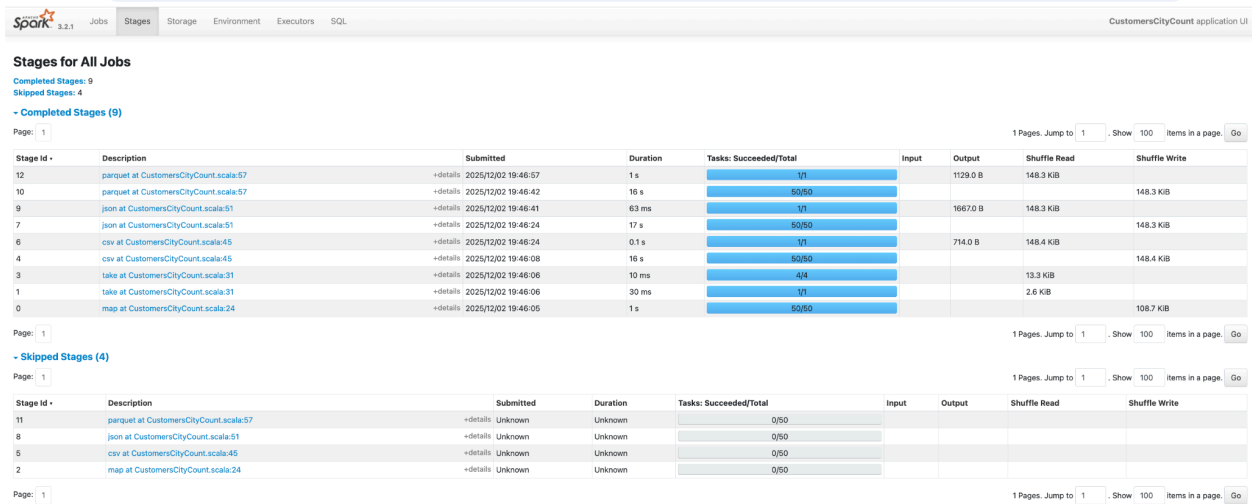
Showing 1 to 1 of 1 entries

Previous1Next

Jobs



Stages



Storage

ID	RDD Name	Storage Level	Cached Partitions	Fraction Cached	Size in Memory	Size on Disk
3	MapPartitionsRDD	Memory Deserialized 1x Replicated	4	8%	672.0 B	0.0 B
5	ShuffledRDD	Memory Deserialized 1x Replicated	4	8%	672.0 B	0.0 B

Stages

Stages

SalesData application UI

Stages for All Jobs

Completed Stages: 8
Skipped Stages: 3

Completed Stages (8)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
10	parquet at SalesData.scala:43	+details 2025/12/02 21:04:32	1 s	1/1		1979.0 B	267.9 KiB	
8	parquet at SalesData.scala:43	+details 2025/12/02 21:04:27	5 s	50/50				267.9 KiB
7	take at SalesData.scala:31	+details 2025/12/02 21:04:25	10 ms	3/3			9.7 KiB	
6	take at SalesData.scala:31	+details 2025/12/02 21:04:25	7 ms	1/1			3.8 KiB	
4	map at SalesData.scala:21	+details 2025/12/02 21:04:20	6 s	50/50				186.3 KiB
3	take at SalesData.scala:30	+details 2025/12/02 21:04:19	0.2 s	3/3			3.8 MiB	
1	take at SalesData.scala:30	+details 2025/12/02 21:04:19	0.1 s	1/1			2.1 MiB	
0	map at SalesData.scala:21	+details 2025/12/02 21:04:14	5 s	50/50				102.8 MiB

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Skipped Stages (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
9	parquet at SalesData.scala:43	+details Unknown	Unknown	0/50				
6	map at SalesData.scala:21	+details Unknown	Unknown	0/50				
2	map at SalesData.scala:21	+details Unknown	Unknown	0/50				

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

3. LOG FILES

Doc Analysis

Time (sec)	Jobs	Stages	Tasks	Shuffle Size	Notes
56.8	1	1	40	425.4 MiB	ERROR TXT WRITE DONE
53.0	1	1	40	65.5 MiB	FULL JSON WRITE DONE
111.2	1	1	81	110.9 KiB	COMPLETED LogFiles

Stages

Stages

LogFiles application UI

Stages for All Jobs

Completed Stages: 2

Completed Stages (2)

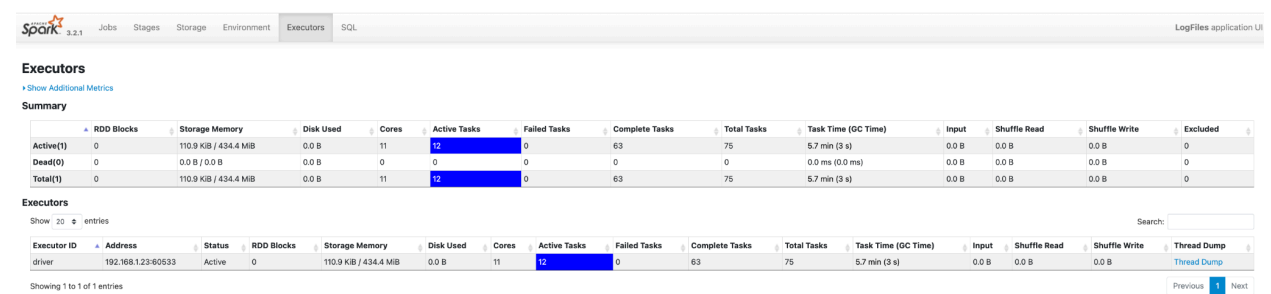
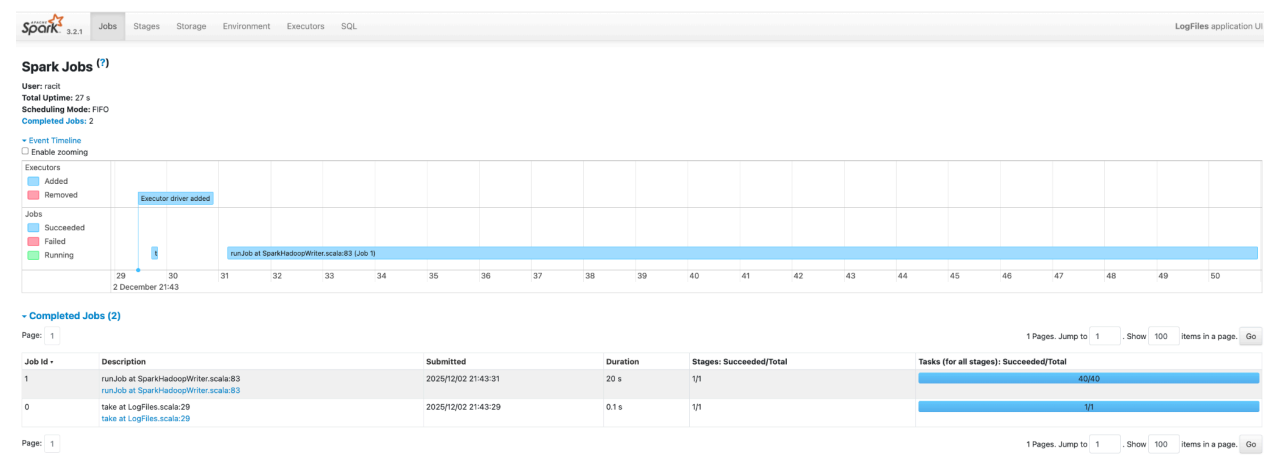
Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
1	runJob at SparkHadoopWriter.scala:83	+details 2025/12/02 21:43:31	20 s	40/40		65.6 MiB		
0	take at LogFiles.scala:29	+details 2025/12/02 21:43:29	45 ms	1/1				

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go



4. PRODUCT CATALOG

Doc Analysis

Time (sec)	Jobs	Stages	Tasks	Shuffle Size	Notes
87.9	1	1	40	71.6 MiB	CSV WRITE DONE
84.3	1	1	40	71.6 MiB	PARQUET WRITE DONE
173.5	1	1	132	71.6 MiB	COMPLETED ProductCatalog

Jobs

Spark

3.2.1

Jobs

Stages

Storage

Environment

Executors

SQL

ProductCatalog application UI

Spark Jobs ⁽⁷⁾

User: racil

Total Uptime: 11 min

Scheduling Mode: FIFO

Completed Jobs: 3

Event Timeline

Enable zooming

Executors

Added

Removed

Jobs

Succeeded

Failed

Running

2 December 22:29

35

40

45

50

55

0

2 December 22:30

5

10

15

20

25

30

Executor driver added

csv at ProductCatalog.scala:32 (Job 0)

csv at ProductCatalog.scala:32 (Job 1)

csv

Completed Jobs (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	csv at ProductCatalog.scala:32 csv at ProductCatalog.scala:32	2025/12/02 22:30:30	0.9 s	1/1 (1 skipped)	12/12 (40 skipped)
1	csv at ProductCatalog.scala:32 csv at ProductCatalog.scala:32	2025/12/02 22:30:00	29 s	1/1	40/40
0	csv at ProductCatalog.scala:32 csv at ProductCatalog.scala:32	2025/12/02 22:29:33	27 s	1/1	40/40

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Stages

Spark

3.2.1

Jobs

Stages

Storage

Environment

Executors

SQL

ProductCatalog application UI

Stages for All Jobs

Completed Stages: 3

Skipped Stages: 1

Completed Stages (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
3	csv at ProductCatalog.scala:32	<details> 2025/12/02 22:30:30	0.9 s	12/12		81.9 MiB	71.6 MiB	
1	csv at ProductCatalog.scala:32	<details> 2025/12/02 22:30:01	29 s	40/40				71.6 MiB
0	csv at ProductCatalog.scala:32	<details> 2025/12/02 22:29:33	27 s	40/40				

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Skipped Stages (1)

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
2	csv at ProductCatalog.scala:32	<details> Unknown	Unknown	0/40				

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Executors

Spark

3.2.1

Jobs

Stages

Storage

Environment

Executors

SQL

ProductCatalog application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	88.6 KiB / 434.4 MiB	0.0 B	11	0	0	132	132	15 min (4 s)	0.0 B	71.6 MiB	71.6 MiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	88.6 KiB / 434.4 MiB	0.0 B	11	0	0	132	132	15 min (4 s)	0.0 B	71.6 MiB	71.6 MiB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:61262	Active	0	88.6 KiB / 434.4 MiB	0.0 B	11	0	0	132	132	15 min (4 s)	0.0 B	71.6 MiB	71.6 MiB	Thread Dump

Show 1 of 1 entries

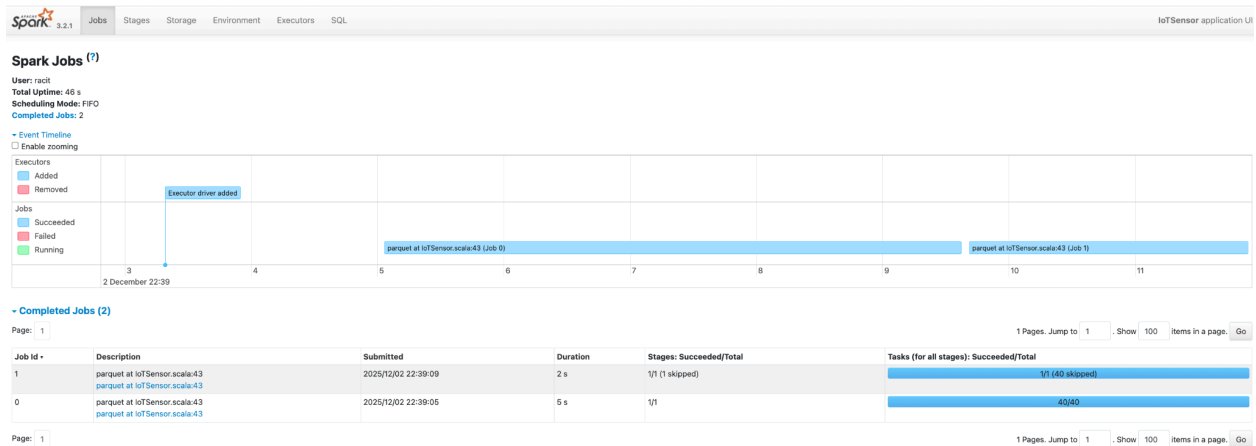
Previous1Next

5. IoT SENSOR DATA

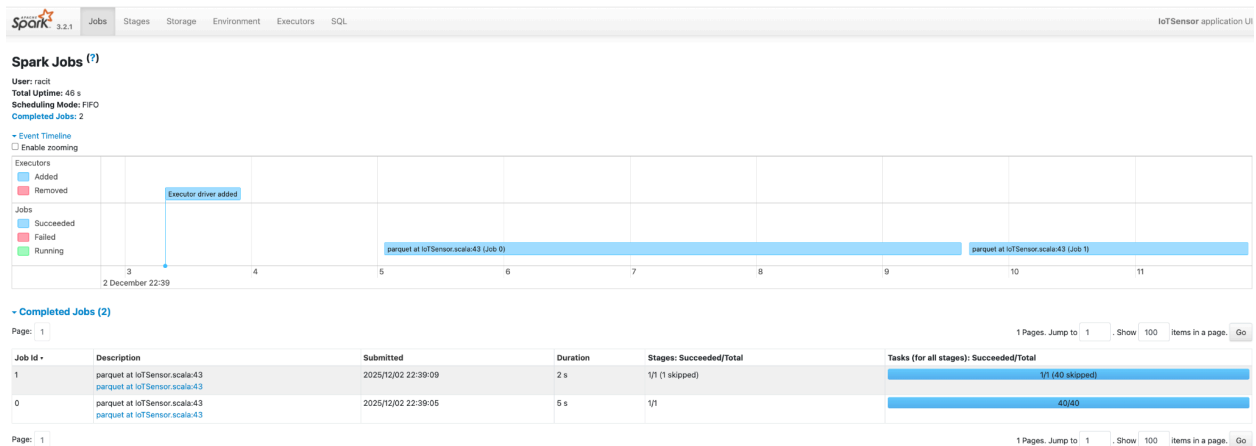
Doc Analysis

Time (sec)	Jobs	Stages	Tasks	Shuffle Size	Notes
35.8	1	1	41	61.6 KiB	PARQUET PARTITIONED WRITE DONE
37.4	1	1	41	61.6 KiB	COMPLETED IoTSensor

Jobs



Stages



Executors

Spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

IoT Sensor application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	103.4 KiB / 434.4 MiB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KiB	61.6 KiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	103.4 KiB / 434.4 MiB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KiB	61.6 KiB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:61480	Active	0	103.4 KiB / 434.4 MiB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KiB	61.6 KiB	Thread Dump

Showing 1 to 1 of 1 entries

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6. SOCIAL MEDIA USERS

Doc Analysis

Time (sec)	Job s	Stage s	Task s	Shuffle Size	Notes
40.9	1	1	40	32.3 KiB	JSON WRITE DONE
42.2	1	1	40	32.3 KiB	COMPLETED SocialMedia (Join triggered)

Jobs

Spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

IoT Sensor application UI

Spark Jobs

User: root

Total Uptime: 46 s

Scheduling Mode: FIFO

Completed Jobs: 2

Event Timeline

Enable zooming

Executors

Added

Removed

Jobs

Succeeded

Failed

Running

Completed Jobs (2)

Page: 1

1 Pages. Jump to: 1. Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
1	parquet at IoT Sensor.scala:43	2025/12/02 22:39:09	2 s	1/1 (1 skipped)	1/1 (40 skipped)
0	parquet at IoT Sensor.scala:43	2025/12/02 22:39:05	5 s	1/1	40/40

Page: 1

1 Pages. Jump to: 1. Show 100 items in a page. Go

Stages

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

IoT Sensor application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(T)	0	103.4 KB / 434.4 MB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KIB	61.6 KIB	0
Dead(T)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(T)	0	103.4 KB / 434.4 MB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KIB	61.6 KIB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:61480	Active	0	103.4 KB / 434.4 MB	0.0 B	11	0	0	41	41	42 s (0.4 s)	0.0 B	61.6 KIB	61.6 KIB	Thread Dump

Showing 1 to 1 of 1 entries

Previous

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Executors

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

SocialMedia application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(T)	0	0.0 B / 434.4 MB	0.0 B	11	0	0	82	82	1.6 min (2 s)	0.0 B	19.2 MIB	19.2 MIB	0
Dead(T)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(T)	0	0.0 B / 434.4 MB	0.0 B	11	0	0	82	82	1.6 min (2 s)	0.0 B	19.2 MIB	19.2 MIB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:61609	Active	0	0.0 B / 434.4 MB	0.0 B	11	0	0	82	82	1.6 min (2 s)	0.0 B	19.2 MIB	19.2 MIB	Thread Dump

Showing 1 to 1 of 1 entries

Previous

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Next

7. FINANCIAL TRANSACTIONS

Doc Analysis

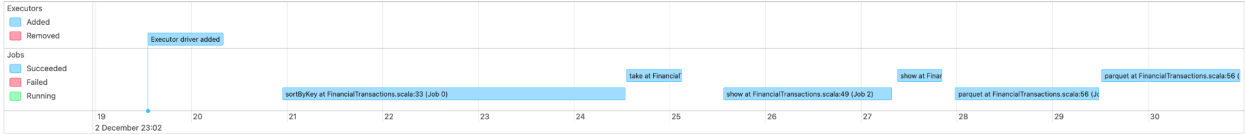
Time (sec)	Job s	Stage s	Task s	Shuffle Size	Notes
33.1	1	1	227	128 MiB	PARQUET WRITE DONE
41.3	1	1	227	100.4 MiB	COMPLETED FinancialTransactions

Jobs

Spark Jobs ⁽⁷⁾

User: root
Total Uptime: 21 s
Scheduling Mode: FIFO
Completed Jobs: 6

Event Timeline
Enable zooming



Completed Jobs (6)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
5	parquet at FinancialTransactions.scala:56 parquet at FinancialTransactions.scala:56	2025/12/02 23:02:29	1 s	2/2 (1 skipped)	13/13 (40 skipped)
4	parquet at FinancialTransactions.scala:56 parquet at FinancialTransactions.scala:56	2025/12/02 23:02:27	2 s	1/1	40/40
3	show at FinancialTransactions.scala:49 show at FinancialTransactions.scala:49	2025/12/02 23:02:27	0.6 s	2/2 (1 skipped)	13/13 (40 skipped)
2	show at FinancialTransactions.scala:49 show at FinancialTransactions.scala:49	2025/12/02 23:02:25	2 s	1/1	40/40
1	take at FinancialTransactions.scala:35 take at FinancialTransactions.scala:35	2025/12/02 23:02:24	0.6 s	2/2 (1 skipped)	41/41 (40 skipped)
0	sortByKey at FinancialTransactions.scala:33 sortByKey at FinancialTransactions.scala:33	2025/12/02 23:02:20	4 s	2/2	80/80

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stages

Stages for All Jobs

Completed Stages: 10
Skipped Stages: 3

Completed Stages (10)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
12	parquet at FinancialTransactions.scala:56	+details 2025/12/02 23:02:29	1 s	1/1		930.0 B	3.2 KIB	
11	parquet at FinancialTransactions.scala:56	+details 2025/12/02 23:02:29	0.3 s	12/12			34.9 MIB	3.2 KIB
9	parquet at FinancialTransactions.scala:56	+details 2025/12/02 23:02:27	2 s	40/40				34.9 MIB
8	show at FinancialTransactions.scala:49	+details 2025/12/02 23:02:27	8 ms	1/1			3.2 KIB	
7	show at FinancialTransactions.scala:49	+details 2025/12/02 23:02:27	0.6 s	12/12			34.9 MIB	3.2 KIB
5	show at FinancialTransactions.scala:49	+details 2025/12/02 23:02:25	2 s	40/40				34.9 MIB
4	take at FinancialTransactions.scala:35	+details 2025/12/02 23:02:25	21 ms	1/1			47.9 KIB	
3	map at FinancialTransactions.scala:32	+details 2025/12/02 23:02:24	0.6 s	40/40			28.1 MIB	1608.6 KIB
1	sortByKey at FinancialTransactions.scala:33	+details 2025/12/02 23:02:23	0.7 s	40/40			28.1 MIB	
0	map at FinancialTransactions.scala:21	+details 2025/12/02 23:02:20	3 s	40/40				28.1 MIB

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Skipped Stages (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
10	parquet at FinancialTransactions.scala:56	+details Unknown	Unknown	0/40				
6	show at FinancialTransactions.scala:49	+details Unknown	Unknown	0/40				
2	map at FinancialTransactions.scala:21	+details Unknown	Unknown	0/40				

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Executors

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	116.9 KIB / 434.4 MIB	0.0 B	11	0	0	227	227	1.5 min (2 s)	0.0 B	128 MIB	100.4 MIB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	116.9 KIB / 434.4 MIB	0.0 B	11	0	0	227	227	1.5 min (2 s)	0.0 B	128 MIB	100.4 MIB	0

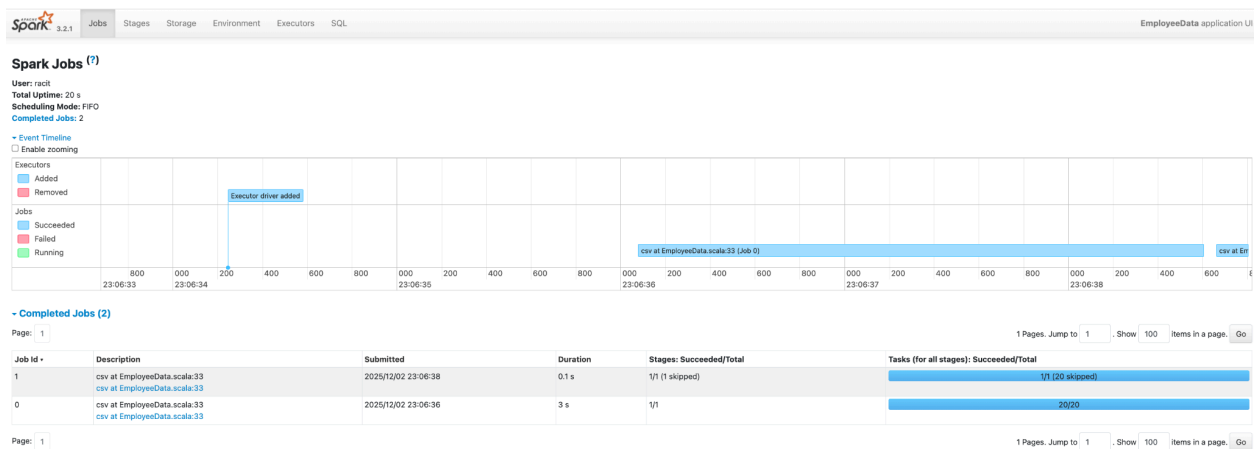
Executors

Show 20 entries															Search:		
Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump		
driver	192.168.1.49:61668	Active	0	116.9 KB / 434.4 MB	0.0 B	11	0	0	227	227	1.5 min (2 s)	0.0 B	128 MB	100.4 MB	Thread Dump		
Showing 1 to 1 of 1 entries																	
															Previous	1	Next

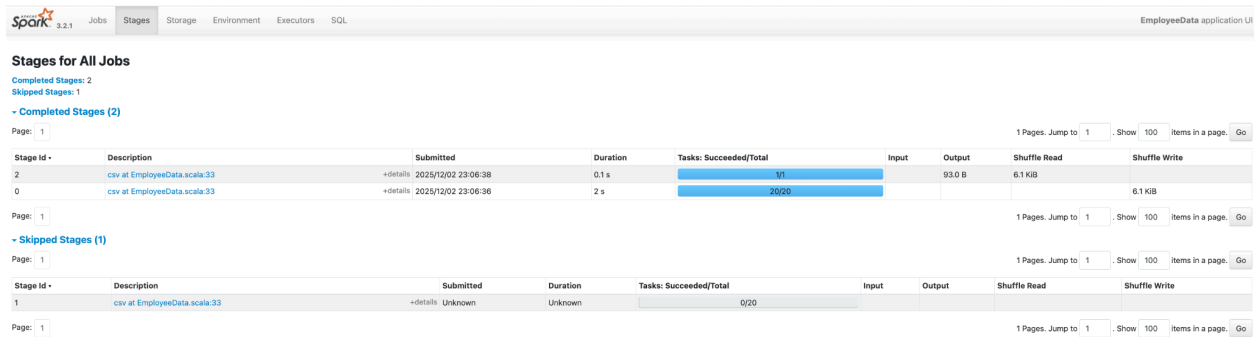
Doc Analysis

Time (sec)	Jobs	Stages	Tasks	Shuffle Size	Notes
3.1	1	1	21	6.1 KiB	CSV WRITE DONE
4.4	1	1	21	6.1 KiB	COMPLETED EmployeeData

Jobs



Stages



Executors

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

EmployeeData application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	103.9 KiB / 434.4 MiB	0.0 B	11	0	0	21	21	25 s (0.3 s)	0.0 B	6.1 KiB	6.1 KiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	103.9 KiB / 434.4 MiB	0.0 B	11	0	0	21	21	25 s (0.3 s)	0.0 B	6.1 KiB	6.1 KiB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:61711	Active	0	103.9 KiB / 434.4 MiB	0.0 B	11	0	0	21	21	25 s (0.3 s)	0.0 B	6.1 KiB	6.1 KiB	Thread Dump

Showing 1 to 1 of 1 entries

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9. STUDENT SCORES

Doc Analysis

Time (sec)	Job s	Stage s	Task s	Shuffle Size	Notes
36.3	1	1	31	19.5 MiB	JSON WRITE DONE
37.6	1	1	31	19.5 MiB	COMPLETED StudentScores

Jobs

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

StudentScores application UI

Spark Jobs ⁽⁷⁾

User: root

Total Uptime: 21 s

Scheduling Mode: FIFO

Completed Jobs: 3

Event Timeline

Enable zooming

Executors

Added

Removed

Jobs

Succeeded

Failed

Running

Completed Jobs (3)

Page: 1

1 Pages. Jump to: 1. Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	json at StudentScores.scala:28 json at StudentScores.scala:28	2025/12/02 23:10:39	1 s	1/1 (1 skipped)	11/11 (20 skipped)
1	json at StudentScores.scala:28 json at StudentScores.scala:28	2025/12/02 23:10:37	3 s	1/1	20/20
0	json at StudentScores.scala:28 json at StudentScores.scala:28	2025/12/02 23:10:34	3 s	1/1	20/20

Page: 1

1 Pages. Jump to: 1. Show 100 items in a page. Go

Stages

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

StudentScores.application UI

Stages for All Jobs

Completed Stages: 3

Skipped Stages: 1

▼ Completed Stages (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
3	jeon at StudentScores.scala:28	+details 2026/12/02 23:10:39	1.0 s	11/11		68.9 MiB	19.5 MiB	
1	jeon at StudentScores.scala:28	+details 2026/12/02 23:10:37	3 s	20/20				19.5 MiB
0	jeon at StudentScores.scala:28	+details 2026/12/02 23:10:34	3 s	20/20				

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

▼ Skipped Stages (1)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
2	jeon at StudentScores.scala:28	+details Unknown	Unknown	0/20				

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Executors

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

StudentScores.application UI

Executors

► Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	0.0 B / 434.4 MiB	0.0 B	11	0	0	51	51	59 s (1 s)	0.0 B	19.5 MiB	19.5 MiB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	0.0 B / 434.4 MiB	0.0 B	11	0	0	51	51	59 s (1 s)	0.0 B	19.5 MiB	19.5 MiB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:01876	Active	0	0.0 B / 434.4 MiB	0.0 B	11	0	0	51	51	59 s (1 s)	0.0 B	19.5 MiB	19.5 MiB	Thread Dump

Showing 1 to 1 of 1 entries

Previous1Next

10. MINI CASE STUDY

Doc Analysis

Time (sec)	Job s	Stage s	Task s	Shuffle Size	Notes
39.4	1	1	—	—	PARQUET WRITE DONE
40.8	1	1	—	—	COMPLETED MiniCaseStudy

Jobs

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

MiniCaseStudy application UI

Spark Jobs (7)

User: racil

Total Uptime: 21 s

Scheduling Mode: FIFO

Completed Jobs: 3

Event Timeline

Enable zooming

Executors

Added

Removed

Jobs

Succeeded

Failed

Running

2 December 23:18

6

7

8

9

10

11

12

13

14

15

16

17

Executor driver added

parquet at MiniCaseStudy.scala:52 (Job 1)

parquet at MiniCaseStudy.scala:52 (Job 0)

parquet at MiniCaseStudy.scala:52 (Job 2)

Completed Jobs (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Job id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	parquet at MiniCaseStudy.scala:52 parquet at MiniCaseStudy.scala:52	2025/12/02 23:18:13	4 s	1/1 (2 skipped)	1/11 (130 skipped)
1	parquet at MiniCaseStudy.scala:52 parquet at MiniCaseStudy.scala:52	2025/12/02 23:18:08	5 s	1/1	80/80
0	parquet at MiniCaseStudy.scala:52 parquet at MiniCaseStudy.scala:52	2025/12/02 23:18:08	4 s	1/1	50/50

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Stages

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

MiniCaseStudy application UI

Stages for All Jobs

Completed Stages: 3

Skipped Stages: 2

Completed Stages (3)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
4	parquet at MiniCaseStudy.scala:52	+details 2025/12/02 23:18:13	4 s	1/11		21.0 MB	78.1 MB	
1	parquet at MiniCaseStudy.scala:52	+details 2025/12/02 23:18:08	2 s	80/80				68.1 MB
0	parquet at MiniCaseStudy.scala:52	+details 2025/12/02 23:18:08	4 s	50/50				10.0 MB

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Skipped Stages (2)

Page: 1

1 Pages. Jump to 1. Show 100 items in a page. Go

Stage id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
3	parquet at MiniCaseStudy.scala:52	+details Unknown	Unknown	0/50				
2	parquet at MiniCaseStudy.scala:52	+details Unknown	Unknown	0/80				

Page: 11 Pages. Jump to 1. Show 100 items in a page. Go

Executors

spark3.2.1

JobsStagesStorageEnvironmentExecutorsSQL

MiniCaseStudy application UI

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Excluded
Active(1)	0	91.1 KB / 434.4 MB	0.0 B	11	0	0	141	141	1.6 min (3 s)	0.0 B	78.1 MB	78.1 MB	0
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B	0
Total(1)	0	91.1 KB / 434.4 MB	0.0 B	11	0	0	141	141	1.6 min (3 s)	0.0 B	78.1 MB	78.1 MB	0

Executors

Show 20 entries

Search:

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump
driver	192.168.1.49:82053	Active	0	91.1 KB / 434.4 MB	0.0 B	11	0	0	141	141	1.6 min (3 s)	0.0 B	78.1 MB	78.1 MB	Thread Dump

Show 1 to 1 of 1 entries

Previous1Next